

KELSO-LONGVIEW, WA, USA

WMO: 727924

Lat: 46.117N

Lon: 122.894W

Elev: 6

StdP: 101.25

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24223

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-4.8	-2.8	-8.9	1.8	-1.4	-6.3	2.2	-0.8	9.7	7.8	8.7	8.1	0.5	140	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	31.3	19.6	28.7	19.0	27.0	18.1	20.6	29.2	19.6	27.6	18.5	25.8	3.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	12.3	24.1	16.3	11.6	22.6	15.7	11.1	21.7	58.6	29.3	55.5	27.4	52.2	25.9	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.6	5.6	-7.9	35.5	2.7	2.3	-9.8	37.2	-11.4	38.5	-12.9	39.8	-14.9	41.5		
			-8.6	22.7	2.6	1.5	-10.5	23.7	-12.0	24.6	-13.5	25.4	-15.4	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	5.1	5.8	8.1	10.2	13.5	16.0	19.0	19.1	16.5	11.6	7.6	4.6
	DBStd	5.90	3.21	2.99	2.75	2.86	3.07	2.97	2.72	2.53	2.78	2.79	3.21	3.33
	HDD10.0	648	154	118	73	30	4	0	0	0	0	16	84	169
	HDD18.3	2629	410	350	318	244	153	83	23	21	68	211	323	426
	CDD10.0	1180	2	2	13	36	113	180	278	282	196	64	12	2
	CDD18.3	118	0	0	0	1	3	13	43	44	14	1	0	0
	CDH23.3	1427	0	0	1	23	88	186	462	450	209	7	0	0
	CDH26.7	479	0	0	0	6	23	62	164	162	62	1	0	0
Wind	WSAvg	2.2	2.5	2.2	2.3	2.1	2.1	2.2	2.4	2.2	1.9	1.9	2.4	2.6
Precipitation	PrecAvg	1177	168	123	115	95	70	52	23	37	55	101	177	181
	PrecMax	1466	346	281	217	183	157	121	82	149	126	201	375	362
	PrecMin	876	7	4	14	31	9	12	0	0	2	7	28	56
	PrecStd	169	84	63	45	38	36	26	20	35	35	52	76	72
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	16.2	22.1	27.0	29.5	32.5	34.5	34.1	32.1	23.9	17.3	13.0
		MCWB	11.3	9.5	12.5	15.5	17.3	20.2	21.5	20.2	18.7	15.5	14.1	11.2
	2%	DB	12.3	13.7	17.7	22.2	26.0	27.8	31.1	31.0	27.9	21.0	15.0	12.0
		MCWB	10.3	9.2	11.2	13.7	16.4	18.2	20.1	19.7	18.2	14.8	12.2	10.5
	5%	DB	11.0	12.3	15.1	18.5	22.5	25.0	27.9	27.8	26.0	18.0	13.1	10.2
		MCWB	9.3	8.8	9.7	11.8	15.0	16.9	19.1	19.0	17.1	13.6	11.0	8.7
	10%	DB	9.1	11.0	12.8	16.0	19.8	22.4	26.0	25.9	22.8	16.3	12.3	8.9
		MCWB	7.6	8.4	9.2	10.5	13.8	15.7	18.0	18.1	16.1	13.0	10.6	7.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	11.7	13.7	16.3	18.9	20.9	22.7	21.7	20.3	17.4	14.9	12.0
		MCDB	13.4	13.8	19.2	24.7	27.8	30.7	32.6	31.3	28.9	20.8	16.7	12.8
	2%	WB	10.6	10.4	11.9	14.0	16.9	18.8	20.7	20.4	18.9	15.6	13.0	10.3
		MCDB	11.5	12.4	15.6	20.2	24.1	27.1	29.3	28.8	26.7	18.9	14.6	11.5
	5%	WB	9.1	9.3	10.8	12.4	15.5	17.3	19.5	19.4	17.7	14.4	11.8	8.9
		MCDB	10.3	11.4	14.2	17.1	21.5	24.3	27.4	27.3	24.3	17.3	12.9	9.9
	10%	WB	7.9	8.1	9.8	11.1	14.1	16.0	18.3	18.3	16.6	13.4	10.7	7.8
		MCDB	9.0	9.8	13.0	14.8	19.0	21.5	25.2	25.0	21.9	16.0	12.0	8.6
Mean Daily Temperature Range	5% DB	MDBR	5.7	7.9	8.9	10.0	10.6	10.5	12.1	12.2	12.0	9.2	6.5	5.3
		MCDBR	7.2	11.4	13.6	16.6	16.9	16.6	17.0	17.1	17.8	13.1	7.9	6.0
		MCWBR	5.1	7.3	7.9	8.9	8.9	8.4	8.2	8.1	8.8	7.7	5.6	4.5
	5% WB	MCDBR	5.7	8.3	11.1	13.8	14.6	14.7	15.5	15.2	15.3	10.1	6.0	4.8
		MCWBR	4.5	5.8	6.8	7.8	8.0	7.9	7.9	7.6	7.9	6.5	4.6	4.1
Clear-Sky Solar Irradiance	taub	0.324	0.316	0.333	0.363	0.378	0.367	0.353	0.358	0.344	0.334	0.324	0.329	
	taud	2.504	2.548	2.509	2.390	2.370	2.419	2.478	2.472	2.510	2.539	2.523	2.435	
	Ebn at Noon	779	870	899	895	888	898	907	890	875	833	772	719	
	Edh at Noon	67	76	91	112	118	113	105	102	90	76	65	64	
All-Sky Solar Radiation	RadAvg	1.10	1.94	2.83	4.14	5.26	5.79	6.53	5.59	4.17	2.40	1.29	0.91	
	RadStd	0.18	0.27	0.37	0.39	0.48	0.56	0.48	0.40	0.37	0.29	0.17	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+25

Nomenclature: See separate page